



SWE 4603

Software Testing and Quality Assurance

Week 3

Prepared By Maliha Noushin Raida, Lecturer, CSE
Islamic University of Technology

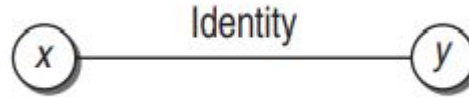
Cause-Effect Graphing Based Testing

- ❖ Like decision tables, cause-effect graphing is another technique for combinations of input conditions.
- ❖ Here, causes are the input conditions and effects are the results of those input conditions.
- ❖ The goal is to reduce the total number of test cases, still achieving the desired application quality by covering the necessary test cases for maximum coverage.
- ❖ A cause is a distinct input condition identified in the problem. Similarly, an effect is an output condition

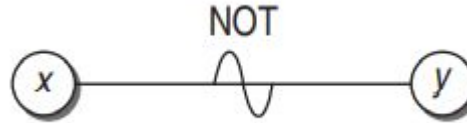
Cause-Effect Graphing Based Testing

Common Notations for Cause Effect Graph:

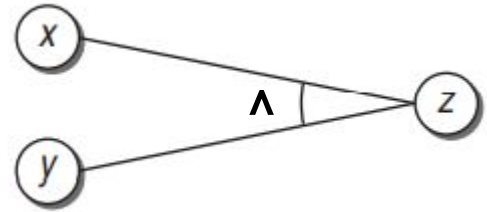
Identity: if x is true, y is true; else y is false.



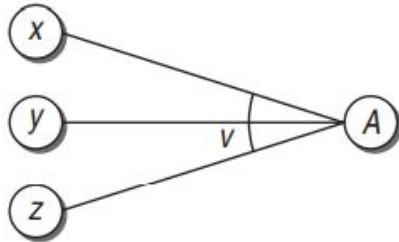
NOT: if x is true, y is false; else y is true.



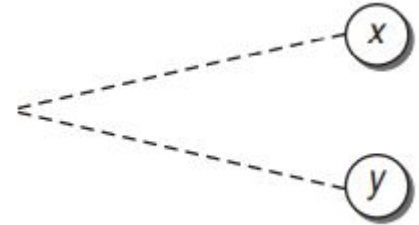
AND: if both x and y are true, z is true; else z is false.



OR: if x or y or z is true, A is true; else A is false.



Mutually Exclusive: One and only one of x and y must be true,



Cause-Effect Graphing Based Testing

Example:

- ❖ The “Print message” is software that reads two characters and, depending on their values, files are updated or messages is printed.
 - The first character must be an “A” or a “B”.
 - The second character must be a digit.
 - If the first character is an “A” or “B” and the second character is a digit, then the file must be updated.
 - If the first character is incorrect (not an “A” or “B”), the message X must be printed.
 - If the second character is incorrect (not a digit), the message Y must be printed.

Cause-Effect Graphing Based Testing

Solution:

The Causes of this situation are:

C1 – First character is A

C2 – First character is B

C3 – the Second character is a digit

The Effects (results) for this situation are:

E1 – Update the file

E2 – Print message “X”

E3 – Print message “Y”

Cause-Effect Graphing Based Testing

Try to figure out:

In this example, let's start with Effect E1.

Effect E1 is for updating the file. The file is updated when:

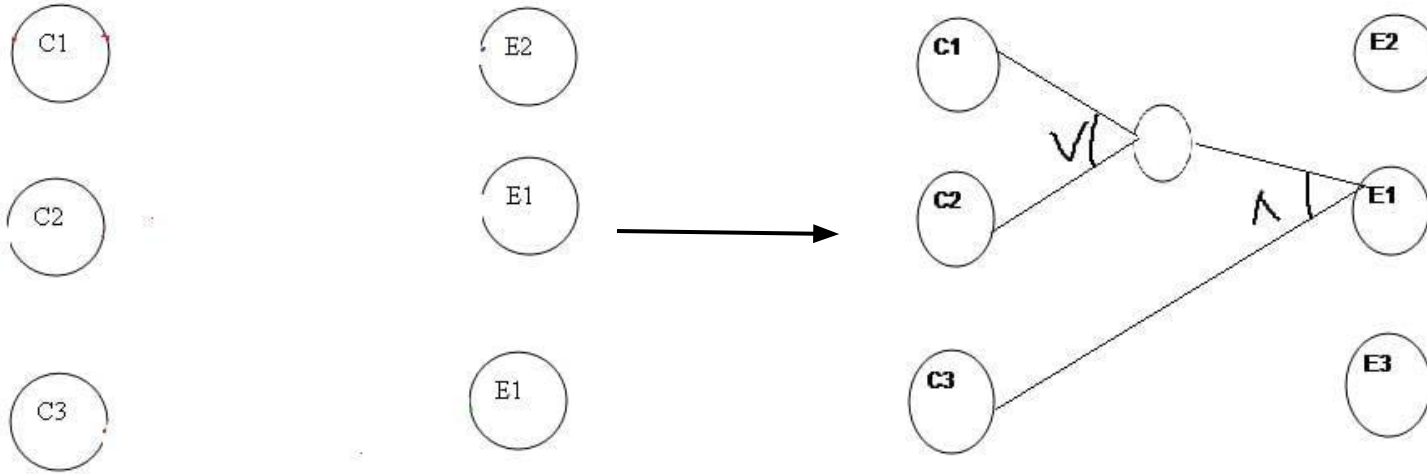
- The first character is "A" and the second character is a digit
- The first character is "B" and the second character is a digit
- The first character can either be "A" or "B" and cannot be both.

Now let's put these 3 points in symbolic form:

For E1 to be true – the following are the causes:

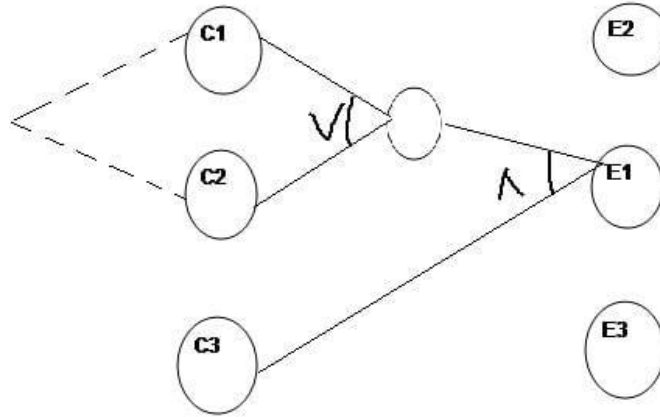
- C1 and C3 should be true
- C2 and C3 should be true
- C1 and C2 cannot be true together. This means C1 and C2 are mutually exclusive.

Cause-Effect Graphing Based Testing



for E1 to be true the condition is $(C1 \vee C2) \wedge C3$

Cause-Effect Graphing Based Testing

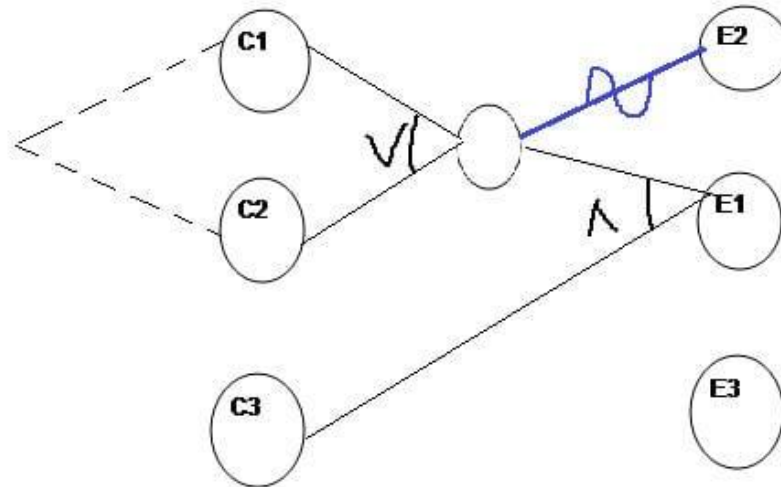


There is a third condition where C1 and C2 are mutually exclusive. So the final graph for effect E1 to be true is shown below:

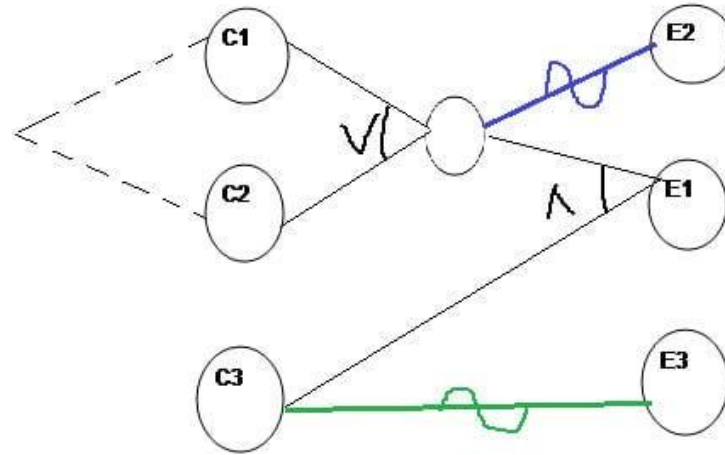
Cause-Effect Graphing Based Testing

E2 states print message "X". Message X will be printed when the First character is neither A nor B.

This means Effect E2 will hold true when either C1 OR C2 is invalid. So the graph for Effect E2 is shown as (In blue line)



Cause-Effect Graphing Based Testing



E3 states print message "Y". Message Y will be printed when the Second character is incorrect.

This means Effect E3 will hold true when C3 is invalid. So the graph for Effect E3 is shown as (In Green line)

Cause-Effect Graphing Based Testing

	R1	R2	R3	R4	R5	R6
C1	T	I	F	I	T	I
C2	I	T	I	F	I	T
C3	T	T	F	T	F	F
E1	X	X				
E2			X	X		
E3					X	X

TC ID	TC Name	Description	Steps	Expected result
TC1	TC1_FileUpdate Scenario1	Validate that system updates the file when first character is A and second character is a digit.	1. Open the application. 2. Enter first character as "A" 3. Enter second character as a digit	File is updated.
TC2	TC2_FileUpdate Scenario2	Validate that system updates the file when first character is B and second character is a digit.	1. Open the application. 2. Enter first character as "B" 3. Enter second character as a digit	File is updated.

Error Guessing

- ❖ Error guessing is the preferred method used when all other methods fail.
- ❖ According to this method, errors or bugs can be guessed which do not fit in any of the earlier defined situations..
- ❖ So test cases are generated for these special cases.
- ❖ this capability comes with years of experience in a particular field of testing.
- ❖ error guessing is an ad hoc approach, based on intuition, experience, knowledge of project, and bug history.

Error Guessing

- ❖ For example, consider the system for calculating the roots of a quadratic
- ❖ equation. Some special cases in this system are as follows:
- ❖ What will happen when $a = 0$? Though, we do consider this case, there are chances that we overlook it while testing, as it has two cases:
 - (i) If $a = 0$ then the equation is no longer quadratic.
 - (ii) For calculation of roots, division is by zero.
- ❖ What will happen when all the inputs are negative?
- ❖ What will happen when the input list is empty?